

QUESTION 1: STUDENTINTERNSHIP DATABASE (1NF → 3NF)

UNF Table

StdID	StdName	DeptID	DeptName	InternshipCompany	Skills	MentorName
101	Hamna	D01	CS	Microsoft	[Python, ML]	Dr. Tayyab
102	Tayyab	D01	CS	Google	[Python, Web Dev, SQL]	Dr. Hamna
103	Areeba	D02	EE	Tesla	[Circuit Design, MATLAB]	Dr. Irfan

Your Task:

1. Convert to **1NF** (one row per student-skill pair)
2. Identify all **Functional Dependencies**
3. Derive **Primary Key**
4. Normalize to **2NF**
5. Normalize to **3NF**
6. Draw dependency diagram if required

QUESTION 2: HOSPITAL TREATMENT DATABASE (1NF → 3NF)

UNF Table

PatientID	PatientName	DoctorID	DoctorName	Treatments	TreatmentDate	RoomNo
P01	Zoya	D01	Dr. Fahad	[MRI, Blood Test]	12-06-2025	R12
P02	Ali	D01	Dr. Fahad	[X-Ray, CT Scan, ECG]	12-06-2025	R12
P03	Mariam	D02	Dr. Sana	[Blood Test, ECG]	13-06-2025	R10

Your Task:

1. Convert to **1NF** by expanding the repeating Treatments field
2. Find **functional dependencies** (Doctor → Room? Patient → Doctor?)
3. Identify **primary keys**
4. Normalize to **2NF** and then to **3NF**
5. Finalize table set